

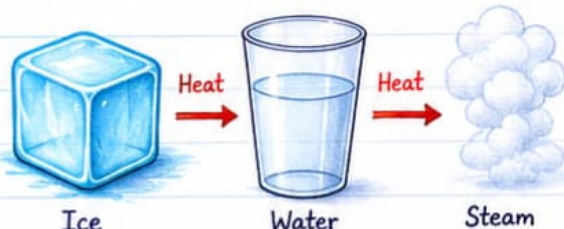
THERMAL PROPERTIES OF MATTER

Key Idea

Heat changes the state, size and behaviour of matter.

1. INTRODUCTION

- All bodies are made up of particles.
- The particles are in continuous random motion.
- Temperature is a measure of the average kinetic energy of particles.
- When heat is supplied to a body, its temperature, state or size may change.



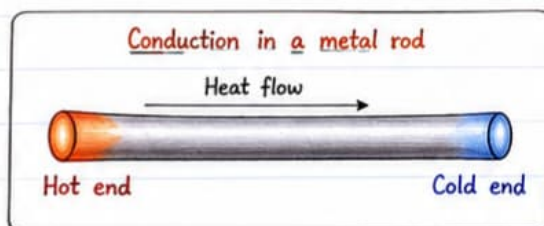
Heat can change the state of matter.

2. MODES OF HEAT TRANSFER

Heat can flow from a body at higher temperature to a body at lower temperature by three modes.

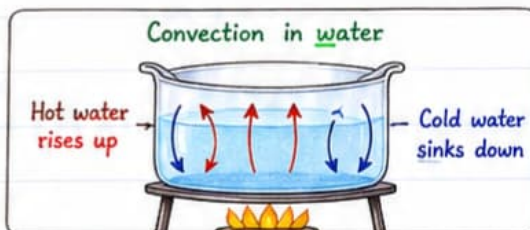
(i) **Conduction**: Transfer of heat through a material without any bulk movement of its particles.

- Example**: Heating one end of a metal rod. The other end becomes hot gradually.



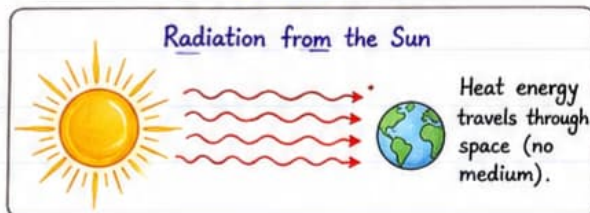
(ii) **Convection**: Transfer of heat through a liquid or gas by the actual movement of the particles.

- Example**: Boiling of water. Hot water rises up and cold water comes down.



(iii) **Radiation**: Transfer of heat in the form of electromagnetic waves. It does not require any medium.

- Example**: Heat from the Sun reaches the Earth through space.

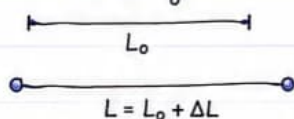


3. THERMAL EXPANSION

- When a body is heated, its size increases. This is called **thermal expansion**.
- It occurs in all **three dimensions** - length, area and volume.

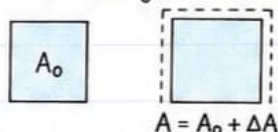
(i) Linear Expansion

Increase in length on heating.



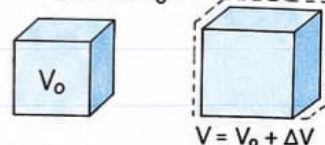
(ii) Areal Expansion

Increase in area on heating.



(iii) Volumetric Expansion

Increase in volume on heating.



Note

- Expansion is more in gases, less in liquids and least in solids.
- On cooling, contraction occurs.



4. SPECIFIC HEAT CAPACITY

- The amount of heat required to raise the temperature of **unit mass** of a substance by 1 K (or 1°C) is called its **specific heat capacity**.
- SI Unit : $\text{J kg}^{-1} \text{K}^{-1}$
- It depends on the nature of the substance and its physical state.
- Water** has a high specific heat capacity. That means it can absorb or release a large amount of heat with a small change in temperature.

Formula

$$Q = mc\Delta T$$

$Q \rightarrow$ Heat absorbed or released (J)

$m \rightarrow$ Mass (kg)

$c \rightarrow$ Specific heat capacity ($\text{J kg}^{-1} \text{K}^{-1}$)

$\Delta T \rightarrow$ Change in temperature (K)

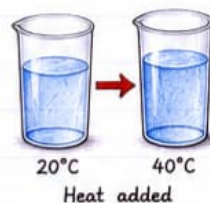
Example

How much heat is required to raise the temperature of 2 kg of water from 20°C to 40°C?

Given: $m = 2 \text{ kg}$, $c = 4.2 \times 10^3 \text{ J kg}^{-1} \text{K}^{-1}$

$$\Delta T = 40 - 20 = 20 \text{ K}$$

$$Q = mc\Delta T = 2 \times 4.2 \times 10^3 \times 20 = 1.68 \times 10^5 \text{ J}$$



5. CHANGE OF STATE (LATENT HEAT)

- During a change of state, **temperature remains constant**. Heat is absorbed or released in the form of **latent heat**.

(i) Latent Heat of Fusion (L_f)

Heat required to change unit mass of a solid into liquid at its melting point without any change in temperature.

$$Q = mL_f$$

(ii) Latent Heat of Vaporisation (L_v)

Heat required to change unit mass of a liquid into gas at its boiling point without any change in temperature.

$$Q = mL_v$$

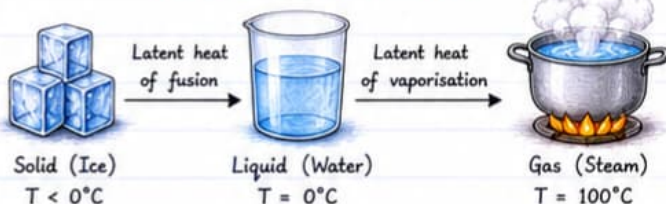
(iii) Latent Heat of Sublimation (L_s)

Heat required to change unit mass of a solid into gas directly without becoming liquid.

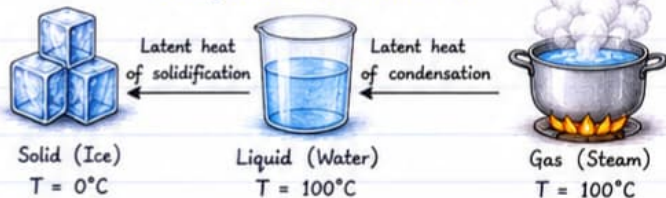
$$Q = mL_s$$

CHANGE OF STATE DIAGRAM

Heating Process (Heat Absorbed)



Cooling Process (Heat Released)



Note

Latent heat **does not** cause any rise in temperature, it only changes the **state** of matter.

6. HEAT CAPACITY

- The amount of heat required to raise the temperature of a body by 1 K is called its **heat capacity**.
- It depends on the mass, specific heat capacity and physical state of the body.

$$C = mc$$

$C \rightarrow$ Heat capacity (J K^{-1})

$m \rightarrow$ Mass (kg)

$c \rightarrow$ Specific heat capacity ($\text{J kg}^{-1} \text{K}^{-1}$)

Example

A copper block of mass 2 kg has a specific heat capacity $0.39 \times 10^3 \text{ J kg}^{-1} \text{K}^{-1}$.

Find its heat capacity.

$$C = mc = 2 \times 0.39 \times 10^3 = 0.78 \times 10^3 \text{ J K}^{-1}$$



7. THERMAL CONDUCTIVITY

- The ability of a material to conduct heat is called **thermal conductivity**.
- SI Unit : $\text{W m}^{-1} \text{K}^{-1}$
- Metals are good conductors of heat (e.g., copper, aluminium).
- Non-metals are poor conductors of heat (e.g., wood, plastic, air).

Good conductors

High thermal conductivity
(Heat flows easily).



Copper rod

Poor conductors

Low thermal conductivity
(Heat flows slowly).



Wooden block

Key Point

Heat causes change in **temperature** (sensible heat) or change in **state** (latent heat).



8. COEFFICIENT OF THERMAL EXPANSION

It is defined as the increase in physical dimension per unit original dimension per unit rise in temperature.

Important

- On heating,
- Length ↑
- Area ↑
- Volume ↑

(ii) Linear Expansion

- If a body of length L_0 at $T_0^\circ\text{C}$ is heated to $T^\circ\text{C}$ and its length becomes L , then increase in length $\Delta L = L - L_0$.
- Coefficient of linear expansion (α) is given by

$$\alpha = \frac{\Delta L}{L_0 \Delta T}$$

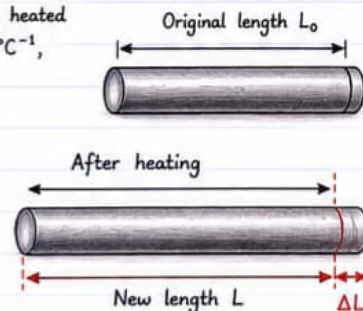
- SI Unit : K^{-1} or $^\circ\text{C}^{-1}$
- Relation : $\Delta L = \alpha L_0 \Delta T$

Example

A steel rod of length 2.0 m at 20°C is heated to 70°C . If α for steel = $1.2 \times 10^{-5} ^\circ\text{C}^{-1}$, find the increase in length.

$$\Delta T = 70 - 20 = 50^\circ\text{C}$$

$$\begin{aligned} \Delta L &= \alpha L_0 \Delta T \\ &= 1.2 \times 10^{-5} \times 2.0 \times 50 \\ &= 1.2 \times 10^{-3} \text{ m} \\ &= \boxed{1.2 \text{ mm}} \end{aligned}$$



(iii) Areal Expansion

- If area A_0 at $T_0^\circ\text{C}$ becomes A at $T^\circ\text{C}$, then $\Delta A = A - A_0$.
- Coefficient of areal expansion (β)

$$\beta = \frac{\Delta A}{A_0 \Delta T}$$

- Relation : $\Delta A = \beta A_0 \Delta T$
- SI Unit : K^{-1} or $^\circ\text{C}^{-1}$

For most solids, $\beta \approx 2\alpha$

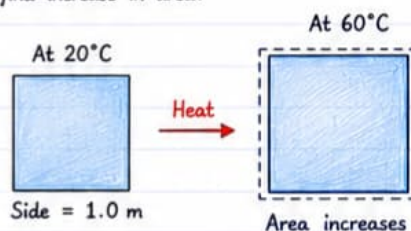
Example

A square metal plate of side 1.0 m at 20°C is heated to 60°C . If $\beta = 2.4 \times 10^{-5} ^\circ\text{C}^{-1}$, find increase in area.

$$\Delta T = 60 - 20 = 40^\circ\text{C}$$

$$A_0 = (1.0)^2 = 1.0 \text{ m}^2$$

$$\begin{aligned} \Delta A &= \beta A_0 \Delta T \\ &= 2.4 \times 10^{-5} \times 1.0 \times 40 \\ &= 9.6 \times 10^{-4} \text{ m}^2 \end{aligned}$$



(iii) Volumetric Expansion

- If volume V_0 at $T_0^\circ\text{C}$ becomes V at $T^\circ\text{C}$, then $\Delta V = V - V_0$.
- Coefficient of volume expansion (γ) is given by

$$\gamma = \frac{\Delta V}{V_0 \Delta T}$$

- Relation : $\Delta V = \gamma V_0 \Delta T$
- SI Unit : K^{-1} or $^\circ\text{C}^{-1}$

For most solids, $\gamma \approx 3\alpha$

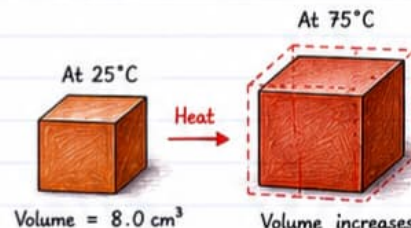
Example

A copper cube of volume 8.0 cm^3 at 25°C is heated to 75°C . If γ for copper = $5.1 \times 10^{-5} ^\circ\text{C}^{-1}$, find increase in volume.

$$\Delta T = 75 - 25 = 50^\circ\text{C}$$

$$V_0 = 8.0 \text{ cm}^3$$

$$\begin{aligned} \Delta V &= \gamma V_0 \Delta T \\ &= 5.1 \times 10^{-5} \times 8.0 \times 50 \\ &= 2.04 \times 10^{-2} \text{ cm}^3 \end{aligned}$$



9. APPLICATIONS OF THERMAL EXPANSION

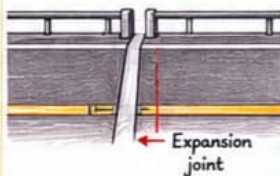
(i) Gaps in Railway Tracks

Small gaps are left between the rails to allow expansion in summer and contraction in winter.



(ii) Expansion Joints

Gaps are provided in bridges and buildings to prevent damage due to expansion.



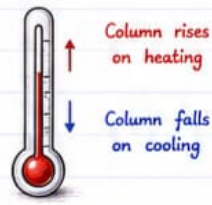
(iii) Tightening of Metal Hoops

A metal hoop is heated and then fitted on a wooden barrel. On cooling, it contracts and becomes tight.



(iv) Thermometer

Works on the principle of thermal expansion of liquids (usually mercury or alcohol).



SUMMARY

- ★ Heat can change the state, size or dimensions of a body.
- ★ Heat transfer occurs by conduction, convection and radiation.
- ★ Specific heat capacity is the heat required to raise temperature of unit mass by 1 K.
- ★ Latent heat is absorbed or released during change of state without change in temperature.
- ★ Heat capacity is the heat required to raise temperature of a body by 1 K.
- ★ Thermal conductivity tells how well a material conducts heat.

Remember

Heat → Motion
Motion → Energy
Energy → Change in state or size



10. COMPARISON & KEY FORMULAS

Type of Expansion	Physical Dimension	Coefficient (Symbol)	Formula	SI Unit	Applies to
Linear	Length (L)	α	$\Delta L = \alpha L_0 \Delta T$	K^{-1} or $^{\circ}C^{-1}$	Rods, wires, rails, etc.
Areal	Area (A)	β	$\Delta A = \beta A_0 \Delta T$	K^{-1} or $^{\circ}C^{-1}$	Plates, sheets, surfaces
Volumetric	Volume (V)	γ	$\Delta V = \gamma V_0 \Delta T$	K^{-1} or $^{\circ}C^{-1}$	Solids, liquids, gases
Specific Heat Capacity	—	c	$Q = m c \Delta T$	$J\ kg^{-1}\ K^{-1}$	All substances
Heat Capacity	—	C	$Q = C \Delta T$	$J\ K^{-1}$	Whole body/system.

RELATIONS BETWEEN COEFFICIENTS

- ★ For most isotropic solids : $\beta = 2\alpha$
 $\gamma = 3\alpha$

Note : These relations are approximate and not valid for all substances.



Remember

For gases at low pressure and moderate temperature, $\gamma \approx 3\alpha$ (ideal gases).

11. EXAMPLE SUMMARY (QUICK RECAP)

(i) Linear Expansion

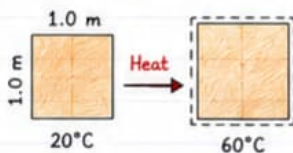
A copper wire of length 1.0 m at $20^{\circ}C$ is heated to $70^{\circ}C$.
 $\alpha = 1.7 \times 10^{-5}\ ^{\circ}C^{-1}$
 Find increase in length.



$$\begin{aligned}\Delta T &= 50^{\circ}C \\ \Delta L &= \alpha L_0 \Delta T \\ &= 1.7 \times 10^{-5} \times 1.0 \times 50 \\ &= 8.5 \times 10^{-4}\ m \\ &= \boxed{0.85\ mm}\end{aligned}$$

(ii) Areal Expansion

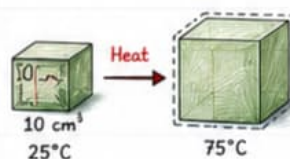
A square plate (side 1.0 m) at $20^{\circ}C$ is heated to $60^{\circ}C$.
 $\beta = 2.4 \times 10^{-5}\ ^{\circ}C^{-1}$
 Find increase in area.



$$\begin{aligned}\Delta T &= 40^{\circ}C \\ A_0 &= (1.0)^2 = 1.0\ m^2 \\ \Delta A &= \beta A_0 \Delta T \\ &= 2.4 \times 10^{-5} \times 1.0 \times 40 \\ &= 9.6 \times 10^{-4}\ m^2\end{aligned}$$

(iii) Volumetric Expansion

A brass cube of volume $10\ cm^3$ at $25^{\circ}C$ is heated to $75^{\circ}C$.
 $\gamma = 5.0 \times 10^{-5}\ ^{\circ}C^{-1}$
 Find increase in volume.



$$\begin{aligned}\Delta T &= 50^{\circ}C \\ \Delta V &= \gamma V_0 \Delta T \\ &= 5.0 \times 10^{-5} \times 10 \times 50 \\ &= 2.5 \times 10^{-2}\ cm^3\end{aligned}$$

(iv) Specific Heat Capacity

How much heat is required to raise the temperature of 3 kg of water from $20^{\circ}C$ to $80^{\circ}C$?
 $c = 4.2 \times 10^3\ J\ kg^{-1}\ K^{-1}$



$$\begin{aligned}\Delta T &= 60^{\circ}C \\ Q &= m c \Delta T \\ &= 3 \times 4.2 \times 10^3 \times 60 \\ &= 7.56 \times 10^5\ J\end{aligned}$$

12. IMPORTANT POINTS

- ★ Thermal expansion is a **general property** of matter.
- ★ Solids expand less, liquids more and gases most on heating.
- ★ Expansion occurs in all **three dimensions**.
- ★ Heat capacity of a body depends on its **mass** and **specific heat capacity**.
- ★ Thermal conductivity determines how **fast heat** flows through a material.

Topper Tip

Remember the concepts with simple diagrams and relate them with daily life examples like railway tracks, bridges, metal lids, thermometers etc.



PRACTICE QUESTIONS (NCERT STYLE)

- Q1. Why do the gaps between rails of a railway track need to be left?
- Q2. A steel rod of length 2.0 m at $10^{\circ}C$ is heated to $60^{\circ}C$. If $\alpha = 1.2 \times 10^{-5}\ ^{\circ}C^{-1}$, calculate the increase in length.
- Q3. A metal plate of area $2\ m^2$ at $20^{\circ}C$ is heated to $80^{\circ}C$. If $\beta = 2.4 \times 10^{-5}\ ^{\circ}C^{-1}$, find the increase in area.

Keep Practicing,
 Keep Learning,
 Stay Curious!



- Q4. A copper cube of volume $27\ cm^3$ at $25^{\circ}C$ is heated to $75^{\circ}C$. If $\gamma = 5.1 \times 10^{-5}\ ^{\circ}C^{-1}$, find the increase in volume.
- Q5. How much heat is required to raise the temperature of 5 kg of aluminium from $25^{\circ}C$ to $75^{\circ}C$? ($c = 900\ J\ kg^{-1}\ K^{-1}$)

★ Heat is the energy that causes change in temperature (sensible heat) or in state (latent heat).

